MULTILINGUAL MULTIMODAL TOURISM AGENT FOR SRI LANKA USING KNOWLEDGE GRAPH REASONING

Project Id: 25-26J-119

Project Proposal Report

Abenayaka T.D

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision

Signature of the supervisor: Date: 29/08/21

ABSTRACT

Tourism in Sri Lanka plays a vital role in the national economy, yet travelers face persistent challenges in accessing reliable, multilingual, and context-aware information scattered across fragmented platforms. This research proposes the development of an intelligent tourism assistant for Sri Lanka powered by four integrated components as Multilingual and Multimodal Language Service, a Sri Lankan Tourism Knowledge Graph with Artificial Intelligence (AI) Reasoning, a Proactive Recommendation and Sustainability Service, and a Multi-Agent Orchestration Layer with Domain Agents.

The core of this work is the Sri Lankan Tourism Knowledge Graph & AI Reasoning Engine, which unifies heterogeneous data sources such as government portals, transport systems, hotel Application Programming Interfaces (API), and cultural event listings into a semantically structured, provenance rich graph. Advanced reasoning capabilities will support multi-hop queries, temporal and event-driven integration, and explanation trails for transparency. Complementing this, the multilingual service will provide Sinhala English code-mixed support, Sinhala OCR, and transliteration for inclusivity, while the recommendation service will re-rank options by combining personalization, sustainability, and forecasting. Finally, a multi-agent orchestration layer will coordinate thin domain agents over the shared KG backbone, ensuring contextual collaboration and fault-tolerant execution.

The novelty of this research lies in introducing the first Sri Lanka specific, event-aware tourism knowledge graph integrated with hybrid reasoning proactive sustainability-aware recommendation, and comparative orchestration strategies. Anticipated outcomes include improved user experience for both local and foreign travelers through context-aware itineraries, support for sustainable tourism practices such as overtourism avoidance, and actionable analytics for government and industry stakeholders. By combining semantic web, AI reasoning, multilingual Natural Language Processing (NLP), and multi-agent coordination, this framework advances the state of intelligent tourism systems and sets a foundation for scalable, sustainable digital tourism infrastructure in Sri Lanka.

Keywords: Tourism Knowledge Graph, AI Reasoning, Multilingual NLP, Sustainability, Multi-Agent Orchestration

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
KG	Knowledge Graph
NLP	Natural Language Processing
NER	Named Entity Recognition
GNN	Graph Neural Network
API	Application Programming Interface
RAG	Retrieval-Augmented Generation
AIL	Agent Interface Layer
SLTDA	Sri Lanka Tourism Development Authority
PLM	Pre-trained language models
TAO	Tourism Analytics Ontology
CI	Continuous Integration
CD	Continuous Deployment

1.INTRODUCTION

Tourism is one of Sri Lanka's most vital economic pillars, contributing significantly to GDP, employment, and international recognition. Yet, tourists both domestic and international struggle with fragmented and unstructured information when planning their journeys. Transportation timetables, visa requirements, hotel reservations, and local events are dispersed across multiple websites and government portals, often inconsistent and difficult to access in real time. The absence of a centralized digital knowledge infrastructure limits the ability of service providers to deliver personalized, reliable, and multilingual assistance.

Knowledge Graphs (KGs) have emerged as powerful technology to model complex, domain-specific knowledge in a structured, interconnected, and machine-readable way. In the tourism sector, KGs have enabled personalized travel recommendations [1], event-driven cultural integration [4], and semantic enrichment of tourism datasets [5]. Despite these advancements, most existing systems are geographically restricted, lack multilingual adaptability, or fail to support dynamic reasoning for real-time decision-making [6]. Addressing these limitations is essential for advancing Sri Lanka's digital tourism ecosystem.

This research proposes the development of a Sri Lankan KG with AI Reasoning and an Agent Interface Layer (AIL), designed to unify heterogeneous tourism datasets government portals, transport systems, accommodation APIs, and event listings into a single, semantically rich structure. By leveraging pre-trained language models for entity extraction and relation linking [2], the system will transform unstructured and semi-structured content into a provenance-rich KG with temporal validity. On top of this, hybrid reasoning modules will support multi-hop queries, contextual inference, geospatial filtering, and missing-link prediction [9], [11]. The AIL will expose standardized query, reasoning, write, and subscription endpoints, enabling downstream services such as recommendation engines and orchestrators to interact with the KG in a protocol-agnostic

way.

The novelty of this research lies in developing the first Sri Lanka–specific, event- and policy-aware Tourism Knowledge Graph that combines multilingual data acquisition, incremental ingestion, semantic reasoning, and explainability. By embedding temporal and contextual layers (festivals, Poya days, seasonal availability, weather), the KG will move beyond static retrieval to deliver evidence-backed, context-aware answers that support intelligent tourism services.

1.1 Background and Literature Review

The role of knowledge graphs in tourism has expanded significantly in the past decade, with research highlighting their use in personalized recommendation systems [1], semantic enrichment of cultural resources [13], and event-driven attraction planning [4]. KGs overcome data fragmentation by representing tourism entities destinations, hotels, attractions, events as nodes, and their relationship's location, timing, availability as edges, enabling structured queries [3], [6].

Early approaches focused on ontology-driven and rule-based models [8], which, while effective for static use cases, lacked scalability and adaptability. With the rise of pre-trained language models (PLMs), entity extraction and relation linking from unstructured text have significantly improved KG construction pipelines [2]. More recent studies emphasize reasoning over KGs, enabling systems to go beyond fact retrieval into inference, predictive link completion, and explanation [9], [11].

In tourism contexts, such reasoning is critical for dynamic changes such as transport delays, hotel availability fluctuations, or last-minute event announcements [12]. Event-centric knowledge graphs have proven their value in supporting attractions recommendation by organizing temporal, contextual, and sentiment-rich data from travelogues [4]. Similarly, cross-field KG construction methods demonstrate how integrating multiple domains such as food, geography, and history enhances the richness of tourism KGs [5]. However, Sri Lanka-specific implementations remain underdeveloped, with little integration across government and private sector sources.

Additionally, approaches such as incremental KG construction [7] provide practical techniques for freshness and change-data capture, but these are not yet operationalized in Sri Lanka's tourism domain. Explainability-oriented frameworks such as Travel Retrieval-Augmented Generation (RAG)s [12] illustrate how KG-anchored reasoning improves reliability and transparency in recommendations, motivating the inclusion of evidence trails and provenance metadata in this research.

This project builds upon these insights by creating a Sri Lanka-focused KG with multilingual ingest, event and policy awareness, hybrid reasoning, and an explainable AIL that serves as the reasoning backbone for downstream systems.

1.2 Research Gap

Although research on tourism knowledge graphs has made significant progress, several key gaps remain unresolved for the Sri Lankan context. These are discussed below with reference to existing literature.

1. Fragmentation of tourism data sources

Most existing systems are developed for specific regions or narrow use cases. For example, some approaches focus on KG-powered recommendation engines [1], while others outline methodologies for constructing tourism KGs [6] or provide landscape reviews [3]. However, these remain limited in geographic coverage and highlight issues of heterogeneity, scalability, and maintenance costs. Incremental KG construction pipelines have been proposed [7], yet they are rarely applied in tourism domains where schedules, bookings, and events change dynamically. In Sri Lanka, tourism data remains scattered across government portals, transport platforms, hotel systems, and event listings, with no unified knowledge graph to capture and synchronize this information.

2. Limited multilingual adaptability

Current tourism KGs are largely monolingual, designed for English or another dominant language. Research has shown that pre-trained language models can extract entities and relations from text across multiple languages [2], and cultural perspectives have been introduced into KG construction [13]. However, there is little evidence of these methods being operationalized for code-mixed settings, such as Sinhala–English, which are common in Sri Lanka. As a result, local users and international tourists alike lack seamless multilingual and code-mixed support in existing solutions.

3. Lack of reasoning capabilities beyond retrieval

Many tourism KGs still operate as static repositories, offering fact retrieval without advanced reasoning. Conversational recommenders that use KGs [10] improve dialogue quality, while interpretability-focused frameworks [11] and retrieval-augmented methods [12] provide grounding and partial explanations. Yet, these do not extend to multi-hop, constraint-aware reasoning under real-world conditions such as *visa eligibility* → *transport availability* → *lodging* → *events*. No Sri Lanka-specific deployment currently demonstrates reasoning pipelines that combine symbolic rules and predictive graph embeddings for complex planning.

4. Underdeveloped event-centric integration

Tourism is inherently dynamic, but most KGs lack temporal and contextual awareness. Event-driven knowledge graphs have been applied to improve attraction recommendations [4] and news-event integration [8], showing that such methods can capture dynamic cultural and seasonal influences. However, these remain limited to localized event sets and do not scale to a national event-policy layer. Sri Lanka has no KG that formally integrates Poya days, cultural festivals, seasonal closures, or weather-driven availability into tourism planning.

5. Lack of explainable, standardized interfaces for integration

Most existing tourism KGs operate as isolated repositories, without exposing structured, explainable interfaces for consumption by downstream systems. While recommendation-

focused KGs [1], [6] emphasize personalization, they rarely provide evidence trails. Retrieval-enhanced methods [12] expose partial supporting paths but are limited to text-based grounding. At present, no Sri Lanka–specific tourism KG provides a standardized AIL that supports reads, writes, reasoning queries, and real-time subscriptions with provenance and typed errors. Without such AIL, orchestrators, recommendation engines, and external applications cannot reliably integrate KG evidence into decision-making.

Table 1.1Existing Research Comparison

	A	B	C	D	E	Our System
Personalized Tourism Recommendation	✓	✓	✓	✗	✗	✓
Multilingual / Code-Mixed Support (English, Sinhala)	✗	✗	✗	✗	✗	✓
Sri Lanka–Specific Data Integration	✗	✗	✗	✓	✗	✓
Event / Dynamic Knowledge Graphs	✗	✓	✗	✗	✗	✓
AI-Based Reasoning (Multi-Hop / Prediction)	✓	✗	✓	✗	✗	✓
Explainable Interfaces + AIL	✗	✗	✗	✗	✓	✓

Research A - Design and Implementation of Personalized Tourism Recommendation System on Basis of Knowledge Graph [1]

Research B - Tourism Event Knowledge Graph for Attractions Recommendation [4]

Research C - CuPe-KG: Cultural and Personalized Knowledge Graph for Tourism Recommendations [13]

Research D - Data-Driven Methodology for Knowledge Graph Generation Within the Tourism Domain [6]

Research E - Advancing multi-agent system with Explainable Interfaces [14]

Addressing these gaps is crucial for advancing Sri Lanka's tourism sector into a digitally empowered ecosystem. By unifying fragmented datasets into a comprehensive Sri Lanka specific Tourism Knowledge Graph, the proposed system will ensure structured, reliable,

and accessible information for all stakeholders. Unlike existing solutions, it will integrate multilingual and code-mixed adaptability, enabling seamless interactions for both local and foreign tourists. Furthermore, the incorporation of hybrid reasoning capabilities will move beyond static retrieval, allowing the system to deliver predictive insights, execute multi-hop queries, and provide constraint-aware decision support.

The introduction of an event- and policy-aware layer will ensure that cultural festivals, Poya days, seasonal activities, and weather-driven conditions are contextually integrated, significantly improving the accuracy and relevance of travel planning. Finally, through a standardized AIL for reads, writes, reasoning queries, and real-time subscriptions, the system will act as a shared knowledge backbone for downstream services, recommendation engines, and orchestrators. Collectively, these contributions highlight the novelty and importance of this project, positioning it as the first Sri Lanka centric, event-aware, multilingual, and reasoning-enabled tourism knowledge graph capable of addressing both current challenges and future demands.

1.2 Research Problem

Tourism has become one of the key drivers of Sri Lanka's economy, yet the digital ecosystem supporting it remains fragmented, inconsistent, and limited in adaptability. Tourists, both local and international, often face challenges in accessing reliable information on accommodation, transportation, visa regulations, and cultural events, as this data is scattered across multiple unstructured government portals and private platforms. Existing tourism knowledge graphs developed globally are typically region-specific and fail to provide a unified, country-wide integration of information [3], resulting in accessibility and decision-making barriers.

Traditional knowledge graph solutions also lack adaptability to Sri Lanka's multilingual context. Current systems are primarily designed for English users, overlooking the need for Sinhala, Tamil, and code-mixed interactions. Although pre-trained language models have demonstrated the ability to support multilingual entity extraction [2], few tourism

systems have operationalized these capabilities, leaving a gap in providing seamless multilingual and code-mixed support for Sri Lanka.

Furthermore, most existing knowledge graph applications function as static repositories with limited reasoning capabilities. While reasoning mechanisms can reveal hidden relationships and enhance information retrieval [9], such methods are largely absent in the tourism domain. Without predictive or multi-hop reasoning, tourists are restricted to basic fact retrieval rather than receiving contextualized, proactive, and constraint-aware recommendations [10], [11], [12].

The dynamic nature of tourism also demands event-driven integration, especially for cultural festivals, Poya days, seasonal closures, and weather-sensitive activities. Although event-based knowledge graphs have been applied in other tourism contexts [4], there is still no Sri Lanka-specific implementation that captures temporal and policy-aware information for real-time planning.

Finally, existing systems lack standardized and explainable AIL that allows reads, writes, reasoning queries, and real-time subscriptions. While some frameworks emphasize multi-agent coordination [14], they are not specialized for tourism and do not expose KG reasoning with provenance and evidence trails. Without such AIL, downstream services, recommendation engines, and orchestrators cannot fully leverage the KG for decision-making.

Research Problem Statement

How can a Sri Lanka–centric, multilingual, and event-aware Tourism Knowledge Graph be developed that unifies fragmented data sources, incorporates hybrid reasoning for multi-hop and predictive queries, and exposes a standardized AIL with explainable evidence trails to support context-aware, real-time, and reliable tourism assistance?

2. OBJECTIVES

2.1 Main Objective

The overarching objective of this research is to design and implement a Sri Lanka–specific Tourism Knowledge Graph with AI Reasoning and an AIL. The system will unify fragmented and heterogeneous data sources including government portals, transport systems, accommodation platforms, and event listings into a semantically structured, provenance-rich knowledge graph. By incorporating multilingual adaptability, event- and policy-awareness, and hybrid reasoning capabilities, the system aims to deliver context-aware, explainable, and real-time decision support for tourism in Sri Lanka.

2.2 Specific Objectives

1) Build a Sri Lanka–Wide Tourism Knowledge Graph

Unify Sri Lanka's fragmented tourism information government notices, visa requirements, train/bus schedules, accommodation inventories, events, and user feedback into a single, governed graph with clean entities and explicit relationships. The objective is to define a national schema (places, transport, events, policies, lodging), implement reliable ingest with temporal validity (valid-from/valid-to), and attach per-assertion provenance and licensing so every answer is auditable. Instead of siloed pages and ad-hoc datasets, this creates the first country-scale, provenance-rich knowledge backbone for Sri Lanka tourism.

2) Operationalize Multilingual, Code-Mixed Knowledge Acquisition

Make the graph natively Multi-lingual (Sinhala, Tamil, English) and code-mix aware from ingestion to query/answer. The objective is to detect language at segment level, normalize scripts and aliases, and extract entities/relations across Sinhala, English so local names (e.g., transliterations, historical toponyms) resolve to the same node. Most systems remain

English-centric; this objective operationalizes multilingual pipelines that reflect how Sri Lankans communicate online.

3) Make the Graph Event- and Policy-Aware for Real-Time Planning

Embed an event-centric layer (who/what/when/where/how) and machine-actionable policy rules (visa/permits/opening constraints) that stay current via change-data capture. The objective is to link cultural festivals, seasonal closures, weather/seasonality, and official rules directly to itineraries, so trip plans adapt when circumstances change. Rather than static attraction facts, the KG becomes a living model of time and policy, returning answers with citations and validity windows (e.g., "this route is feasible on these dates, visa-eligible under conditions X").

4) Provide Hybrid AI Reasoning for Decisions

Go beyond retrieval by combining symbolic multi-hop reasoning (e.g., visa → transport → lodging → events) with graph learning (embeddings/link-prediction) and constraint checks (time, budget, capacity, season). The objective is to produce feasible, ranked plans with explanations (paths, rules, citations) and to fill data gaps predictively when sources are incomplete. Novelty: this fuses deterministic reasoning (transparent, checkable) with predictive signals (adaptive, robust), delivering proactive, explainable decisions rather than static lists.

5) Develop an Agent Interface Layer

Another important objective is to develop an AIL that will make the knowledge graph easy to use for other systems and services. The AIL will provide clear and well-documented APIs for four main tasks: querying, writing, subscribing, and explaining. Through the query APIs (e.g., /kg/query, /kg/reason), applications can search the knowledge graph or request travel plans based on multiple conditions. The write API (e.g., /kg/write) will allow safe updates to the graph while keeping track of sources and time validity. The subscribed API (e.g., /kg/subscribe) will send live updates such as transport

delays, price changes, or event cancellations. Finally, the explain API (e.g., /kg/explain) will give evidence and reasoning paths so users can see why a particular answer or plan was generated. With these features, the AIL will ensure that the knowledge graph is not just a static data store but a real-time, explainable service that can be directly used by recommendation systems, mobile apps, and orchestration layers in the tourism domain.

3. METHODOLOGY

The proposed system, Sri Lankan Tourism Knowledge Graph with AI Reasoning and an AIL, is designed to unify fragmented tourism information, provide intelligent multilingual adaptability, and deliver real-time, explainable decision support. To achieve these goals, the project leverages knowledge graph construction, hybrid AI reasoning, large-scale data integration, and explainable interfaces. This ensures a scalable, sustainable, and context-aware tourism solution for both local and international travelers.

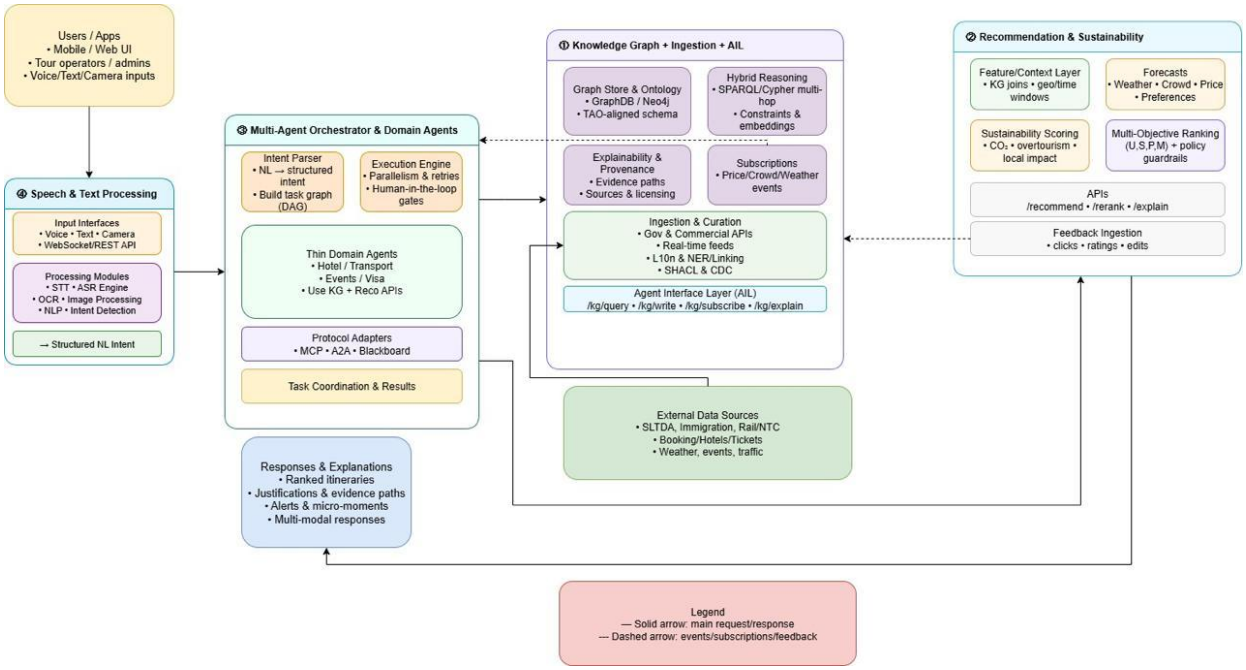

Figure 1Overall System Diagram

As illustrated in Figure 3.1, the system will be developed using a modular microservice architecture to ensure scalability, flexibility, and fault tolerance. This design allows independent services for ingestion, reasoning, and API delivery, while maintaining interoperability across components. The core infrastructure will be centered on a graph database for storing and reasoning over structured tourism knowledge, with additional relational and NoSQL stores for complementary data. The backend will run on a cloud-native containerized platform (Docker + Kubernetes), ensuring resilience, observability, and ease of integration with external services.

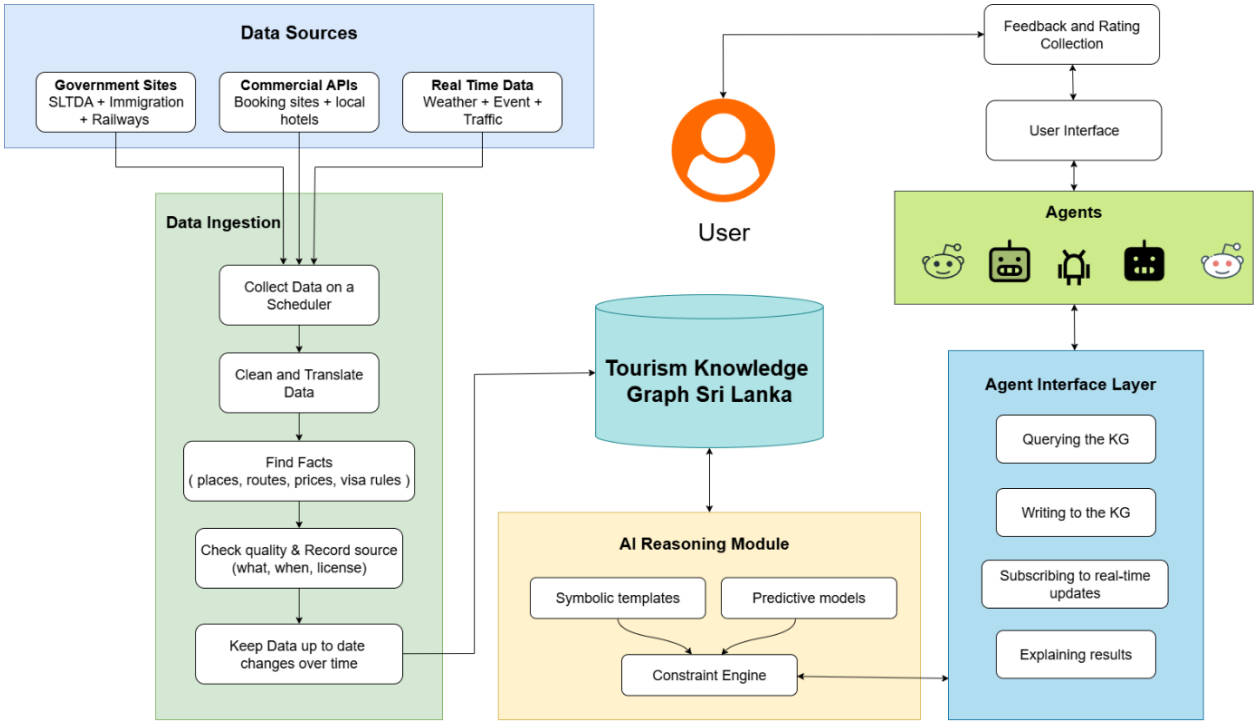

Figure 3.2Component Diagram

The first stage involves acquiring data from heterogeneous sources such as the Sri Lanka Tourism Development Authority (SLTDA), immigration portals, railway and bus reservation systems, hotel booking platforms, and event listings. Public APIs and scraping pipelines will be used to gather structured, semi-structured, and unstructured content. To support inclusivity, multilingual preprocessing will be applied using transformer-based models, ensuring that Sinhala, Tamil, and English text is normalized and linked consistently. The curated data will be mapped to a domain-specific schema using RML/YARRRML mappings, enabling change-data capture and incremental updates.

Once the data is collected, the next step is to construct and enrich the knowledge graph. Named Entity Recognition (NER) and relation extraction will be used to identify tourism-specific entities and their associations. Entity linking will ensure consistency across sources and alignments with external references such as Wikidata and GeoNames. The KG will be modeled using a modular ontology inspired by the Tourism Analytics Ontology (TAO), extended with Sri Lanka–specific classes such as transport routes, Poya days, and event entities. The KG will be stored in Neo4j/GraphDB, supporting efficient traversal, geospatial queries (GeoSPARQL), and semantic reasoning. SHACL validation will be applied to ensure data quality and schema consistency.

The reasoning layer combines symbolic and sub-symbolic methods to enhance decision support. Symbolic reasoning will be implemented using SPARQL and Cypher queries for multi-hop traversals and constraint checks. Sub-symbolic reasoning will use graph embeddings and Graph Neural Networks (GNNs) to support predictive tasks such as missing link prediction, clustering of related entities, and semantic similarity search. This hybrid reasoning approach ensures both deterministic, explainable outputs and adaptive, predictive insights, making the system robust against incomplete or evolving data.

A central novelty of this research is the AIL, which exposes the knowledge graph as an interactive reasoning service. The AIL provides standardized, versioned APIs for

- Querying to access facts, geospatial results, or constraint-aware plans.
- Writing with provenance checks and temporal metadata for safe updates.
- Subscriptions to push real-time updates such as transport delays, price changes, or event cancellations.
- Explanations to return evidence trails, rules fired, and provenance details supporting any answer.

This interface ensures interoperability with external systems, allowing recommendation services, mobile applications, and orchestration layers to integrate seamlessly with the KG. Unlike static repositories, the AIL transforms the KG into a real-time, explainable, and trustworthy decision-support backbone for Sri Lanka's tourism sector.

User queries and feedback, such as itinerary refinements or booking confirmations, will be reintegrated into the KG. This feedback-driven cycle ensures the system evolves in line with changing traveler preferences, seasonal demand patterns, and new events. Incremental ingestion methods [7] will ensure that updates are efficient, minimizing computation and maintaining data freshness.

Technologies

- Graph Databases – Neo4j, Graph DB
- NLP & Data Processing – Transformers spaCy, NLTK
- Reasoning & Inference – Graph Neural Networks (PyTorch Geometric, DGL), Knowledge Graph Embeddings
- API & Data Integration – FastAPI
- Storage & Caching – MongoDB and Redis
- Deployment & Backend – Python 3.10+, Docker, Kubernetes

Algorithms

- NER
- Relation Extraction Algorithms
- Entity Linking Algorithms

Other

- Version Management – GitHub
- Containerization & Orchestration – Docker, Kubernetes
- Monitoring & Logging – Prometheus, Grafana
- Continuous Integration/Deployment (CI/CD) – Jenkins

The design and implementation of the Sri Lankan Tourism Knowledge Graph with AI Reasoning and AIL will be planned, developed, and delivered according to the allocated project timeline and task management plan. The main responsibility of this component is to unify fragmented tourism datasets, construct a provenance-rich knowledge graph, integrate multilingual and event-aware layers, and provide explainable reasoning services through standardized APIs.

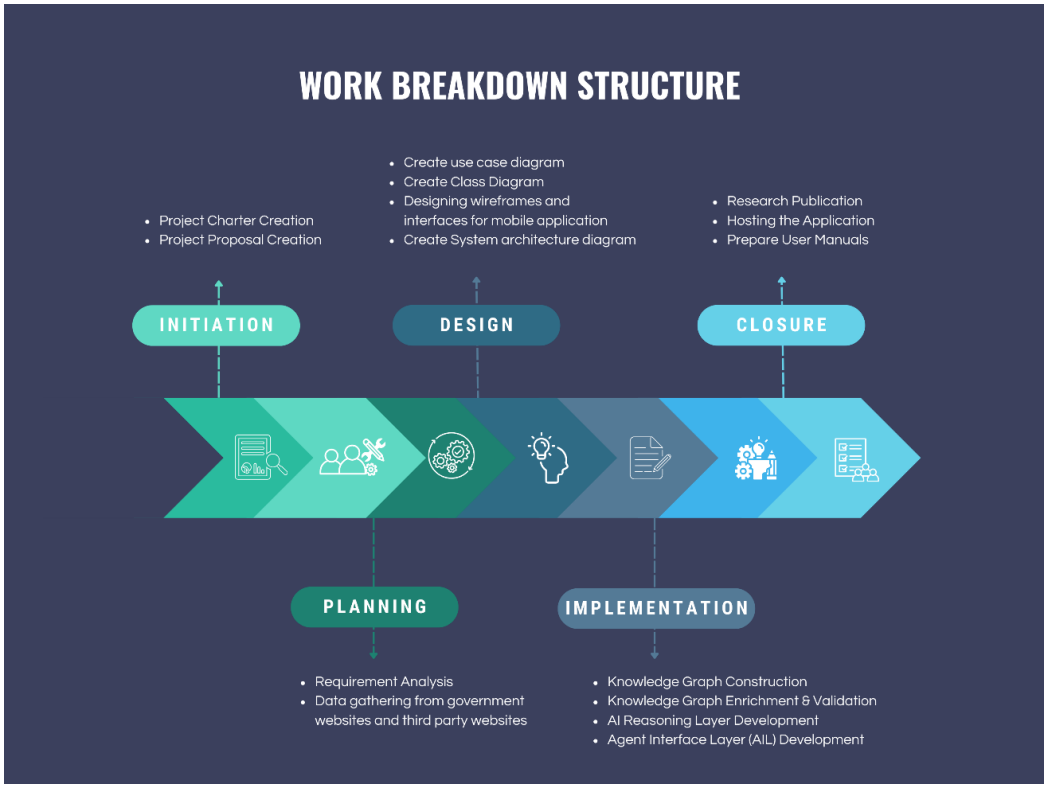

Figure 3.3Work Breakdown Structure

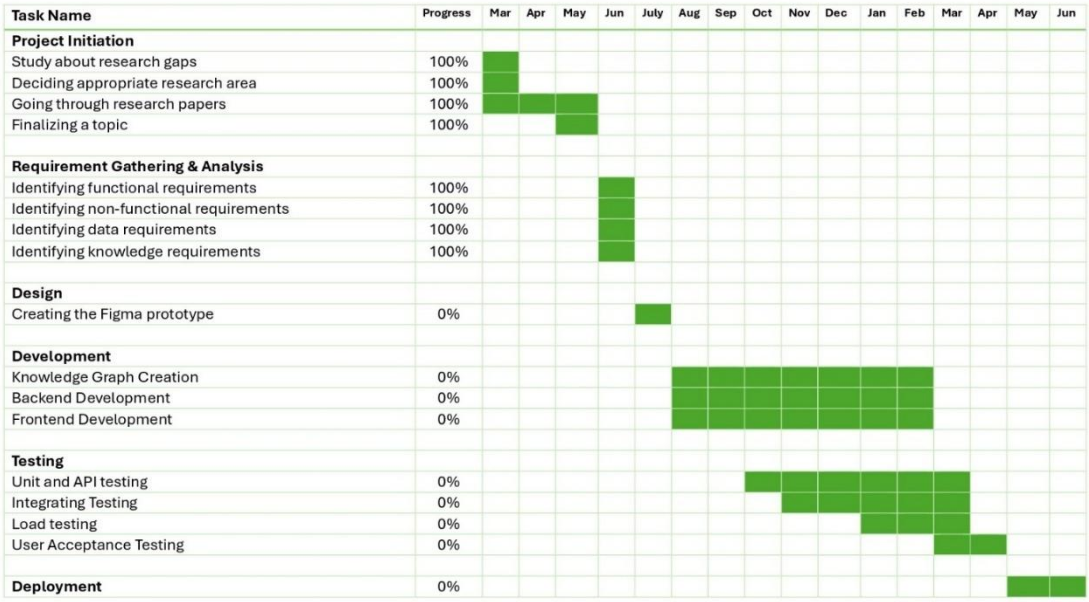

Figure 3.4Gantt Chart

4.PROJECT REQUIREMENTS

The project will commence with a comprehensive requirements-gathering phase, involving collaboration with stakeholders such as the SLTDA, transport authorities, hotel associations, cultural event organizers, and potential travelers. The aim is to define the specific needs for building a unified Sri Lanka-focused Tourism Knowledge Graph with AI reasoning and a standardized AIL. Based on these discussions, the following requirements have been identified

4.1 Functional Requirements

- **Knowledge Graph Construction** – The system must collect, clean, and integrate heterogeneous data sources (government portals, transport schedules, hotel booking APIs, event calendars, and weather feeds) into a structured and provenance-rich knowledge graph.
- **Entity Linking and Relationship Modeling** – The system must extract and normalize entities such as destinations, hotels, cultural events, transport routes, and policies, linking them through semantic and temporal relationships.
- **Reasoning and Multi-Hop Query Support** – The system must support hybrid reasoning, combining symbolic queries and predictive embeddings, to infer hidden relationships and execute multi-hop queries.
- **Event- and Policy-Centric Integration** – The knowledge graph must incorporate temporal and contextual data such as Poya days, festivals, seasonal closures, and visa rules, enabling real-time, context-aware itinerary planning.
- **Agent Interface Layer** – The system must provide a standardized interface for querying, reasoning, writing, subscribing, and explaining. APIs should include /kg/query, /kg/reason, /kg/write, /kg/subscribe, and /kg/explain, with provenance and error handling to ensure explainability and trust.

4.2 Non-Functional Requirements

- **Performance** – The system must support low-latency queries and reasoning to provide real-time responses.
- **Scalability** – The knowledge graph and AIL must scale horizontally to accommodate expanding data sources and increasing user queries.
- **Security** – All data must be stored and transmitted securely, with compliance to data protection standards ensuring proper authentication and authorization.
- **Reliability** – The system must maintain high availability and resilience, especially during peak tourism seasons, with minimal downtime.
- **Usability** – APIs and query interfaces must be intuitive and easy to integrate into conversational agents, recommendation engines, and mobile applications.
- **Explainability** – All reasoning outputs must be accompanied by evidence trails, provenance information, and justification paths for transparency.

4.3 System Requirements

- **Backend Infrastructure** – Cloud-based microservice architecture deployed on Kubernetes with Docker containers for modularity and resilience.
- **Graph Database** – Neo4j or GraphDB for storing entities, relationships, geospatial data, and reasoning queries.
- **APIs** – REST and WebSocket APIs exposed through the AIL for querying, writing, subscribing, and explanations.
- **Data Sources** – Government websites (SLTDA, Immigration, Railway, NTC), third-party APIs (hotels, transport, events, weather), and curated user input.
- **Validation and Mapping** – Use RML for data mappings and SHACL for schema validation to ensure data quality and integrity.

4.4 User Requirements

- Tourists – Should be able to access accurate, multilingual, and context-aware tourism information through conversational agents powered by the knowledge graph.
- Service Providers – Hotels, transport operators, and event organizers should benefit from seamless integration into the system, with real-time updates reflected in the knowledge graph.
- Government Authorities – Should gain access to anonymized analytics for tourism planning, policymaking, and event management.

5.PERSONAL AND FACILITIES

Facilitators:

• Dr. Lakmini Abeywardhena – Sri Lanka institute of Information Technology (SLIIT)

• Dr. Nathali Silva - Sri Lanka institute of Information Technology (SLIIT)

6. BUDGET AND BUDGET JUSTIFICATION

The proposed system has been planned with a lean but realistic budget, making use of free tiers where possible, while allocating essential costs for cloud services, APIs, and mobile deployment. The emphasis is on using Azure cloud services for backend hosting, storage, and speech capabilities, combined with OpenAI API keys for multi-agent reasoning and natural language tasks. In addition, a Flutter-based mobile application will be developed as the final deliverable, requiring costs for developer accounts on app stores. Hardware requirements such as microphones are also included to support data collection and testing. A buffer allocation has been added to account for unforeseen expenses during integration and field validation.

Table 6.1Budget Table

Description	**Estimated Cost**	**Notes**
Azure Cloud Services (VM hosting, Blob Storage, App Services, Speech-to-Text)	LKR 3000 / monthly	Backend hosting, data storage, STT/OCR services
OpenAI API Keys	LKR 3000	For embeddings, orchestration, reasoning tasks
Microphones for Voice Data Collection	LKR 10,000	For Sinhala–English code-mixed dataset collection
Mobile Testing Device	LKR 20,000	For app testing and evaluation
Miscellaneous / Buffer	LKR 10,000	Documentation, SD cards, adapters, printing

Total Budget = LKR 46,000

Budget Justification

Azure Cloud Services

Azure has been selected as the primary cloud provider due to its generous free tiers and reliable integration of storage, compute, and AI services. Virtual Machines and App Services will host the backend and APIs, while Blob Storage will hold datasets and graph exports. Azure Speech-to-Text will support Sinhala–English transcription at competitive rates, making it essential for the multilingual component.

OpenAI API Keys

A developer-tier plan will be used for embedding generation, agent orchestration, and conversational reasoning tasks. This ensures that the system can perform advanced semantic queries and multi-agent coordination effectively.

Flutter Mobile Application

The final deliverable will be a cross-platform mobile application built using Flutter, allowing deployment on both Android and iOS. Publishing costs include the Apple Developer Program and Google Play Developer registration. These are unavoidable costs to ensure the app can be distributed officially and maintained with regular updates.

Microphones & Mobile Device

A high-quality microphone (~LKR 10,000) is required for collecting Sinhala–English speech data to fine-tune STT components. Additionally, a mid-range Android device (~LKR 20,000) is budgeted for testing the mobile application in real-world conditions.

Miscellaneous Costs

A buffer of LKR 5,000 is included for documentation, printing, SD cards, adapters, and other unforeseen requirements during development and evaluation.

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